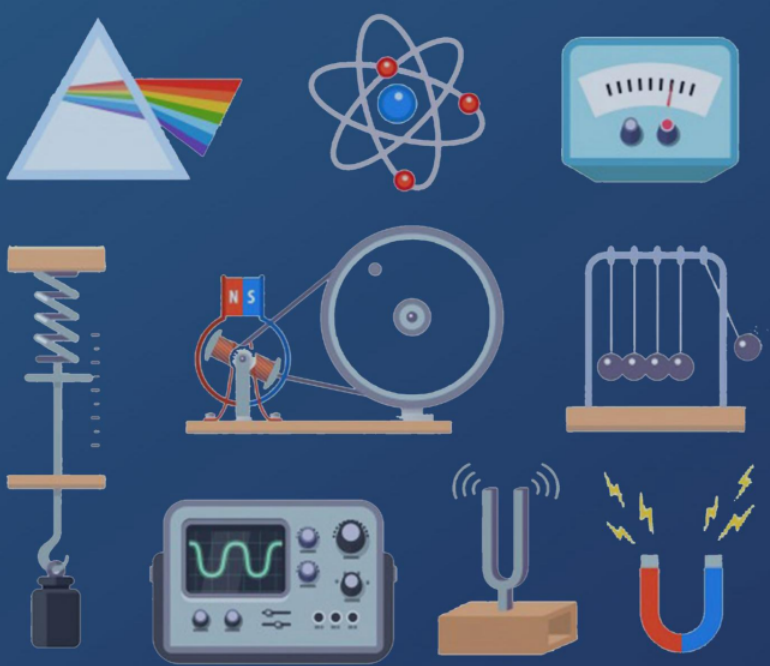
PHYSICS

Thermal Expansion



SHEET

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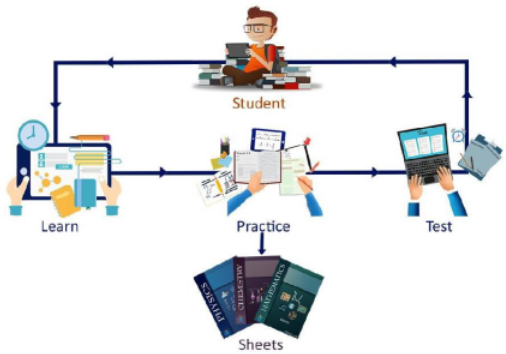
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Thermal Expansion-Introduction	1	7,8	1	
2	Coefficients of Expansion	2-5	1-5	2-4	1,10,11
3	Concepts Expansion	6,7	6		
4	Thermal Stress	8		5	2
5	After Full Chapter			6,7	3-9, 12

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Thermal Expansion-Introduction

Q1. The coefficient of linear expansion of copper is 17×10^{-6} $(°C)^{-1}$. A copper statue is 93 m tall on the summer morning of temperature 25°C. What is maximum order of increase in magnitude of the height in statue (maximum temperature of day is 45°C)

(A) 0.1 mm (B) 1 mm

(C) 10 mm (D) 100 mm

Coefficients of Expansion

Q2. The volume of a metal sphere increases by 0.24% when its temperature is raised by 40°C. The coefficient of linear expansion of the metal is .....°C

(A) 2×10^{-5} (B) 6×10^{-5}

(C) 2.1×10^{-5} (D) 1.2×10^{-5}

Q3. When a copper ball is heated, the largest percentage increase will occur in its

(A) Diameter (B) Area

(C) Volume (D) Density

Q4. A thin copper wire of length L increase in length by 1% when heated from temperature T_1 to T_2. What is the percentage change in area when a thin copper plate having dimensions $2L \times L$ is heated from T_1 to T_2?

(A) 1% (B) 2%

(C) 3% (D) 4%

Q5. If the length of a cylinder on heating increases by 2%, the area of its base will increase by

(A) 0.5% (B) 2%

(C) 1% (D) 4%

Concepts Expansion

Q6. A brass disc fits simply in a hole of a steel plate. The disc from the hole can be loosened if the system.

(A) First heated then cooled

(B) First cooled then heated

(C) Is heated

(D) Is cooled

Q7. Density of substance at 0°C is 10 gm/cc and at 100°C, its density is 9.7 gm/cc. The coefficient of linear expansion of the substance will be :-

(A) 10^2 °C^{-1}

(B) 10^{-2} °C^{-1}

(C) 10^{-3} °C^{-1}

(D) 10^{-4} °C^{-1}

Thermal Stress

Q8. If a bimetallic strip is heated, it will

(A) bend towards the metal with lower thermal expansion coefficient.

(B) bend towards the metal with higher thermal expansion coefficient.

(C) twist itself into helix.

(D) have no bending.

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Ratio among linear expansion coefficient (α), area expansion coefficient (β) and volume expansion coefficient (γ) is -

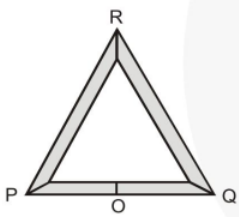
(A) 1 : 2 : 3

(B) 3 : 2 : 1

(C) 4 : 3 : 2

(D) none of these

Q2. Three rods of equal length ℓ are joined to form an equilateral triangle PQR. O is the mid point of PQ. Distance OR remains same for small change in temperature. Coeffcient of linear expansion for PR and RQ is same i.e. α_2 but that for PQ is α_1. Then



(A) $\alpha_2 = 3\alpha_1$ (B) $\alpha_1 = 4\alpha_2$

(C) $\alpha_1 = 3\alpha_2$ (D) $\alpha_2 = 4\alpha_1$

Q3. The coefficient of linear expansion of crystal in one direction is α_1 and that in every direction perpendicular to it is α_2,. The coefficient of cubical expansion is.

(A) $\alpha_1 + \alpha_2$

(B) $2\alpha_1 + \alpha_2$

(C) $\alpha_1 + 2\alpha_2$

(D) None of these

Q4. When the temperature of a copper coin is raised by 80°C, its diameter increases by 0.2%.

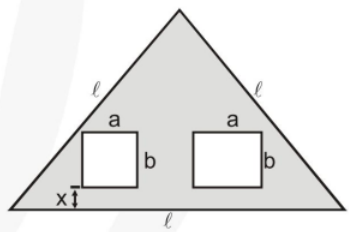
(A) Percentage rise in the area of a face is 0.4 %

(B) Percentage rise in the thickness is 0.4 %

(C) Percentage rise in the volume is 0.6 %

(D) Coefficient of linear expansion of copper is $0.25 \times 10^{-4}\ C^{\circ\ -1}$.

Q5. A triangular plate has two cavities, one square and one rectangular as shown in figure. The plate is heated.



(A) a increase, b decrease

(B) a and b both increase

(C) a and b increase, x and ℓ decrease

(D) a, b, x and ℓ all increase

Q6. Water has maximum density at

(A) 0°C (B) 32°F

(C) - 4°C (D) 4°C

SUBJECTIVE TYPE

Q7. A rod of length 2 m at 0°C and having expansion coefficient

$\alpha = (3x + 2) \times 10^{-6}\ °C^{-1}$,

where x is the distance (in cm) from one end of rod. The length of rod at 20°C is :

Q8. A hole is drilled in a copper sheet. The diameter of the hole is 4.24 cm at 27.0 °C. What is the change in the diameter of the hole when the sheet is heated to 227 °C ?

Coefficient of linear expansion of copper $= 1.70 \times 10^{-5}\ °C^{-1}$.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A gas follows VT^2 = const. Its volume expansion coefficient will be :-

(A) $\frac{2}{T}$ (B) $-\frac{2}{T}$

(C) $\frac{3}{T}$ (D) $-\frac{3}{T}$

Q2. Which of the following is true about coefficient of linear expansion for anisotropic solids?

(A) $\beta_{xy} = \alpha_x + \alpha_y$ (B) $\beta_{xz} = \alpha_x + \alpha_z$

(C) $\gamma = \alpha_x + \alpha_y + \alpha_z$ (D) All of these

Q3. Two rods one of aluminium of length l_1 having coefficient of linear expansion α_a and other steel of length l_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the rods is same on variation of temperature. Then the value of $\frac{l_1}{l_1+l_2}$ is -

(A) $\frac{\alpha_s}{\alpha_a+\alpha_s}$ (B) $\frac{\alpha_s}{\alpha_a-\alpha_s}$

(C) $\frac{\alpha_a+\alpha_s}{\alpha_s}$ (D) None of these

Q4. If two rods of length L and 2L having coefficients of linear expansion α and 2α respectively are connected so that total length becomes 3L, the average coefficient of linear expansion of the composition rod equals :

(A) $\frac{3}{2}\alpha$ (B) $\frac{5}{2}\alpha$

(C) $\frac{5}{3}\alpha$ (D) none of these

Q5. A rod of length 2 m is at a temperature of 20°C. Find the free expansion of the rod, if the temperature is increased to 50°C, then find stress produced when the rod is fully prevented to expand.

($Y = 2 \times 10^{11}$ N/m^2 ; $\alpha = 15 \times 10^{-6}$/°C)

(A) 9×10^7 N/m^2 (B) 3×10^7 N/m^2

(C) 6×10^7 N/m^2 (D) None of these

Q6. A wire of length L_0 is supplied heat to raise its temperature by T. If γ is the coefficient of volume expansion of the wire and Y is Young's modulus of the wire then the energy density stored in the wire is

(A) $\frac{1}{2}\gamma^2T^2Y$ (B) $\frac{1}{3}\gamma^2T^2Y^3$

(C) $\frac{1}{18}\frac{\gamma^2T^2}{Y}$ (D) $\frac{1}{18}\gamma^2T^2Y$

Q7. A rod of length 20 cm is made of metal. It expands by 0.075cm when its temperature is raised from 0°C to 100°C. Another rod of a different metal B having the same length expand by 0.045 cm for the same change in temperature. A third rod of the same length is composed of two parts, one of metal A and the other of metal B. This rod expands by 0.060 cm for the same change in temperature. The portion made of metal A has the length :

(A) 20 cm (B) 10 cm

(C) 15 cm (D) 18 cm

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The volume of the bulb of a mercury thermometer at 0°C is V_0 and cross section of the capillary is A_0. The coefficient of linear expansion of glass is α_g per °C and the cubical expansion of mercury γ_m per °C. If the mercury just fills the bulb at 0°C, what is the length of mercury column in capillary at T°C.

(A) $\frac{V_0T(\gamma_m+3\alpha_g)}{A_0(1+2\alpha_gT)}$ (B) $\frac{V_0T(\gamma_m-3\alpha_g)}{A_0(1+2\alpha_gT)}$

(C) $\frac{V_0T(\gamma_m+2\alpha_g)}{A_0(1+3\alpha_gT)}$ (D) $\frac{V_0T(\gamma_m-2\alpha_g)}{A_0(1+3\alpha_gT)}$

Q2. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficient of linear expansion of the two metals are α_C and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R. Then R is

(A) proportional to ΔT

(B) inversely proportional to ΔT

(C) proportional to $|\alpha_B-\alpha_C|$

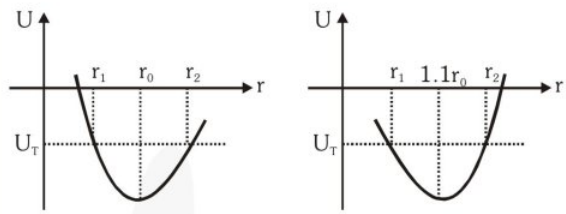
(D) inversely proportional to $|\alpha_B-\alpha_C|$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 3 to 5

Consider a hypothetical situation where we are comparing the properties of two crystals made of atom A and atom B. Potential energy (U) v/s interatomic separation (r) graph for atom A and atom B is shown in figure (i) and (ii) and respectively.



Q3. Choose correct statement :

(A) Volume of A and B expand on heating

(B) Volume of A and B contract on heating

(C) A expands on heating and B contracts on heating

(D) A contracts on heating and B expands on heating

Q4. When we heat the crystal of either atoms, the atom undergo oscillation. Choose correct statement for atoms of crystal A.

(A) Their equilibrium position remains unchanged but average separation decreases

(B) Their equilibrium position remains unchanged but average separation increases

(C) Their separation at equilibrium position as well as average separation increases

(D) Their separation at equilibrium position decreases but average separation increases

Q5. It is seen that the potential energy can reach a maximum value of U_T at temperature T = 10K. If r_1 and r_2 are 0.9999 r_0 and 1.0003 r_0 for atoms of crystal A, its approximate coefficient of linear expansion can be :-

(A) 4×10^{-5} /K (B) 1×10^{-5} /K

(C) 2×10^{-5} /K (D) 3×10^{-5} /K

Paragraph for Question No. 6 to 9

Most substances contract on freezing. However, water does not belong to this category. We know that water expands on freezing. Further, coefficient of volume expansion of water in the temperature range from 0°C to 4°C is negative and above 4°C it is positive. This behaviour of water shapes the freezing of lakes as the atmospheric temperature goes down and it is still above 4°C.

Q6. As the atmospheric temperature goes down, but it is still above 4°C

(A) Cooled water at the surface flows downward because of its greater density.

(B) Cooled water at the surface does not flow downward and remains at the surface because its smaller density.

(C) Cooled water at the surface, through it remains on the surface because of its smaller density, will conduct heat from the interior to the atmosphere.

(D) Cooled water at the surface flows to the bottom because of its smaller density.

Q7. As the atmospheric temperature goes below 4°C.

(A) Cooled water at the surface flows downward because of its greater density.

(B) Cooled water at the surface does not flow downward and remains at the surface because of its smaller density.

(C) Cooled water at the surface downward because of its smaller density.

(D) Temperature of water in the lake reduces with depth.

Q8. If the atmospheric temperature is below 0°C and ice begins to form at t = 0, thickness of ice sheet formed up to a time 't' will be directly proportional to -

(A) t^4 (B) t^2 (C) t (D) $t^{1/2}$

Q9. If the atmospheric temperature is below 0°C

(A) Ice will form from the bottom upward and the plants and animals in the lake will be displaced to the upper part of the lake.

(B) Ice will form in a random manner throughout the volume of the lake and with the passage of time, different segments of ice will join together to result in a collective ice mass

(C) Ice will form from the surface downward and plant and animal life will survive in the water beneath.

(D) Water in the lake does not freeze. In fact, water in the atmosphere freezes and fall into the lake and floats on the surface of lake as ice.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q10. A pendulum clock is 5 sec fast at temperature of 15°C and 10 sec slow at a temperature of 30°C. At what temperature does it give the correct time :-

SUBJECTIVE TYPE

Q11. A clock pendulum made of invar has a period of 0.5 sec at 20°C. If the clock is used in a climate where average temperature is 30°C approximately. How much fast or slow will the clock run in 10^6 sec.

($\alpha_{invar}=1\times10^{-6}/°C$)

Q12. A rod AB of length 'ℓ' is pivoted at an end A and freely rotated in a horizontal plane at an angular speed ω about a vertical axis passing through A. If co-efficient of linear expansion of material of rod is α, find the percentage change in its angular velocity if temperature of system is increased by ΔT.

Answer Key

Ex-1							
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
C	A	C	B	D	D	D	A

Ex-1A

1. A **2.** B **3.** C **4.** A,C,D **5.** B,D

6. D **7.** 2.0002 m **8.** 1.44×10^{-2} cm

Ex-2						
1	**2**	**3**	**4**	**5**	**6**	**7**
B	D	A	C	A	D	B

Ex-2A

1. B **2.** B,D **3.** C **4.** B **5.** C

6. A **7.** B **8.** D **9.** C **10.** 20°C

11. 5 sec slow **12.** $200\alpha\Delta T$